

PART 1: SHORT ANSWER

1. A Database Management System (DBMS) is a dedicated software system designed to store, retrieve, and manage data efficiently. Built-in mechanisms for data integrity, concurrency control, security, and crash recovery.

A traditional file system stores data in isolated files, lacking centralized control, which often leads to data redundancy, inconsistency, and difficulty in querying.

An airline reservation system is a scenario where a DBMS is clearly the better choice. In this environment, hundreds of users are attempting to book the same seats simultaneously. A DBMS handles concurrency control to ensure that two people cannot book the same seat at the exact same millisecond.

2. A strong entity has its own primary key. A weak entity does not possess its own primary key and its existence is entirely dependent on a strong entity. The primary key of a weak entity is formed by combining the primary key of its identifying strong entity with its own partial key (discriminator).

3. It should be modeled as a multivalued attribute. The scenario explicitly states that a student may have multiple phone numbers. A simple attribute can only hold a single atomic value, and a composite attribute is used to break a single concept down into smaller sub-parts. A multivalued attribute is specifically designed to store multiple values of the same type for a single entity record.

4. A super key is any set of attributes that uniquely identifies a row in a table. A candidate key is a *minimal* super key, contains no extraneous attributes.

Candidate keys for Student(roll no, email, name, branch): roll no and email

Super keys for students: (roll no, name) and (email, branch).

5. Referential Integrity Constraint. It protects against orphan records and ensures data consistency by guaranteeing that a foreign key value in one table must have a matching primary key value in the referenced table. The DBMS will reject the insert operation and return an error.

6. A junction table. Its primary key is a composite key consisting of the foreign keys that reference the primary keys of the participating entities. Relational databases require tables to be in at least First Normal Form (1NF), meaning every column must hold an atomic (single) value.

PART 2

Part A: Customer (Primary Key) → A single customer can place multiple orders, but each individual order is placed by exactly one customer

Restaurant (Primary Key) → Each order is associated with exactly one restaurant, but a single restaurant can receive and process multiple orders over time.

Order (Primary Key) → A restaurant can offer multiple menu items, but in this specific schema context, a specific menu item record.

Item (Primary Key) → An order may contain multiple menu items, and a specific menu item can appear in multiple different orders placed by various customers.

Part B:

One Multivalued Attribute: Dietary restrictions

A single customer might have multiple dietary restrictions simultaneously (e.g., Vegan, Gluten-Free). Since multiple independent values of the same nature apply to one entity instance, it defines as a multivalued attribute.

One Composite Attribute: Delivery address

An address is logically a single concept but can be subdivided into smaller parts (Street, City, State) to allow the database to easily query customers by specific regions or cities.

Part C:

Primary keys: OrderID, ItemID

Foreign keys: order(orderID), MenuItem(ItemID).

Additional Attributes: Quantity(say)

PART 3

